

Supplementary material to “The Lasso Distribution: Properties, Sampling Methods, and Applications in Bayesian Lasso Regression”

by Mohammad Javad Davoudabadi, Jonathon Tidswell, Samuel Muller, Garth Tarr
and John T. Ormerod

A Derivation of the full conditional for each β_j

Consider $p(\mathbf{y}|\boldsymbol{\beta}, \boldsymbol{\eta})$, $p(\boldsymbol{\beta}|\boldsymbol{\eta})$, $p(\boldsymbol{\eta})$, $p(\sigma^2)$, and $p(\lambda^2)$ are the likelihood function, and the priors of $\boldsymbol{\beta}$, $\boldsymbol{\eta}$, σ^2 , and λ^2 , respectively. The log of the full joint distribution is proportional to

$$\begin{aligned} \log p(\mathbf{y}, \boldsymbol{\beta}, \sigma^2, \lambda^2, \boldsymbol{\eta}) &= \log (p(\mathbf{y}|\boldsymbol{\beta}, \boldsymbol{\eta})p(\boldsymbol{\beta}|\boldsymbol{\eta})p(\boldsymbol{\eta})p(\sigma^2)p(\lambda^2)) \\ &\propto -\frac{1}{2\sigma^2} \left[\mathbf{y}^T \mathbf{y} - 2\mathbf{y}^T \mathbf{X}^T \boldsymbol{\beta} + \boldsymbol{\beta}^T \mathbf{X}^T \mathbf{X} \boldsymbol{\beta} + \lambda^2 \boldsymbol{\beta}^T \mathbf{A} \boldsymbol{\beta} \right] \\ &\quad - \frac{\sum_{i=1}^p \eta_i}{2} - 2 \sum_{i=1}^p \eta_i - \sum_{i=1}^p \frac{1}{2\eta_i} + \text{constant}; \end{aligned}$$

where $\boldsymbol{\eta} = (\eta_1, \dots, \eta_p)$ and $\mathbf{A} = \text{diag}(\boldsymbol{\eta}^{-1})$. After integrating out $\boldsymbol{\eta}$ from the above proportion, the marginal distribution of β_j is as follows

$$\log p(\beta_j | \text{rest}) \propto -\frac{\|\mathbf{X}_j\|_2^2}{2\sigma^2} \beta_j^2 + \frac{\mathbf{X}_j^T (\mathbf{y} - \mathbf{X}_{-j} \boldsymbol{\beta}_{-j})}{\sigma^2} \beta_j - \frac{\lambda^2}{\sigma} |\beta_j|;$$

which is a univariate Lasso distribution with parameters $a = \frac{\|\mathbf{X}_j\|_2^2}{\sigma^2}$, $b = \frac{\mathbf{X}_j^T (\mathbf{y} - \mathbf{X}_{-j} \boldsymbol{\beta}_{-j})}{\sigma^2}$, and $c = \frac{\lambda^2}{\sigma}$.

B Derivation of the distributional properties of the Lasso distribution

If $X \sim \text{Lasso}(a, b, c)$ with then it has density given by

$$p(X, a, b, c) = Z^{-1} \exp \left(-\frac{1}{2}aX^2 + bX - c|X| \right)$$

where Z is the normalizing constant. Then

$$\begin{aligned} Z(a, b, c) &= \int_{-\infty}^{\infty} \exp \left[-\frac{1}{2}aX^2 + bX - c|X| \right] dX \\ &= \int_0^{\infty} \exp \left[-\frac{1}{2}aX^2 + (b-c)X \right] dx + \int_{-\infty}^0 \exp \left[-\frac{1}{2}aX^2 + (b+c)X \right] dX \\ &= \int_0^{\infty} \exp \left[-\frac{(X-\mu_1)^2}{2\sigma^2} + \frac{\mu_1^2}{2\sigma^2} \right] dx + \int_{-\infty}^0 \exp \left[-\frac{(X-\mu_2)^2}{2\sigma^2} + \frac{\mu_2^2}{2\sigma^2} \right] dX \\ &= \sqrt{2\pi\sigma^2} \left[\exp \left\{ \frac{\mu_1^2}{2\sigma^2} \right\} \int_0^{\infty} \phi(x; \mu_1, \sigma^2) dx + \exp \left\{ \frac{\mu_2^2}{2\sigma^2} \right\} \int_{-\infty}^0 \phi(x; \mu_2, \sigma^2) dx \right] \\ &= \sqrt{2\pi\sigma^2} \left[\exp \left\{ \frac{\mu_1^2}{2\sigma^2} \right\} \{1 - \Phi(-\mu_1/\sigma)\} + \exp \left\{ \frac{\mu_2^2}{2\sigma^2} \right\} \{\Phi(-\mu_2/\sigma)\} \right] \\ &= \sqrt{2\pi\sigma^2} \left[\exp \left(\frac{\mu_1^2}{2\sigma^2} \right) \Phi \left(\frac{\mu_1}{\sigma} \right) + \exp \left(\frac{\mu_2^2}{2\sigma^2} \right) \Phi \left(\frac{-\mu_2}{\sigma} \right) \right] \\ &= \sigma \left[\frac{\Phi(\mu_1/\sigma)}{\phi(\mu_1/\sigma)} + \frac{\Phi(-\mu_2/\sigma)}{\phi(-\mu_2/\sigma)} \right] \end{aligned}$$

where $\mu_1 = (b - c)/a$, $\mu_2 = (b + c)/a$ and $\sigma^2 = 1/a$.

The MGF of the Lasso distribution is

$$\begin{aligned}
M(t) &= \mathbb{E}(\exp(tX)) = Z(a, b, c)^{-1} \int \exp(tX) \exp\left(\frac{-a}{2}X^2 + bX - c|X|\right) dX \\
&= Z(a, b, c)^{-1} \int \exp\left(\frac{-a}{2}X^2 + (b + t)X - c|X|\right) dX \\
&= \frac{Z(a, b + t, c)}{Z(a, b, c)} Z(a, b + t, c)^{-1} \int \exp\left(\frac{-a}{2}X^2 + (b + t)X - c|X|\right) dX \\
&= \frac{Z(a, b + t, c)}{Z(a, b, c)}.
\end{aligned}$$

The moments of the Lasso distribution are:

$$\begin{aligned}
\mathbb{E}(X^r) &= Z^{-1} \int_{-\infty}^{\infty} X^r \exp\left[-\frac{1}{2}aX^2 + bX - c|X|\right] dX \\
&= Z^{-1} \int_0^{\infty} X^r \exp\left[-\frac{1}{2}aX^2 + (b - c)X\right] dX + \int_{-\infty}^0 X^r \exp\left[-\frac{1}{2}aX^2 + (b + c)X\right] dX \\
&= Z^{-1} \sqrt{2\pi\sigma^2} \exp\left(\frac{\mu_1^2}{2\sigma^2}\right) \int_0^{\infty} x^r \phi(x; \mu_1, \sigma^2) dx + \sqrt{2\pi\sigma^2} \exp\left(\frac{\mu_2^2}{2\sigma^2}\right) \int_{-\infty}^0 x^r \phi(x; \mu_2, \sigma^2) dx \\
&= \frac{\sigma}{Z} \left[\frac{\Phi(\mu_1/\sigma)}{\phi(\mu_1/\sigma)} \frac{\int_0^{\infty} X^r \phi(x; \mu_1, \sigma^2) dx}{\Phi(\mu_1/\sigma)} + \frac{\Phi(-\mu_2/\sigma)}{\phi(-\mu_2/\sigma)} \frac{\int_{-\infty}^0 X^r \phi(x; \mu_2, \sigma^2) dx}{\Phi(\mu_2/\sigma)} \right] \\
&= \frac{\sigma}{Z} \left[\frac{\Phi(\mu_1/\sigma)}{\phi(\mu_1/\sigma)} \mathbb{E}(A^r) + \frac{\Phi(-\mu_2/\sigma)}{\phi(-\mu_2/\sigma)} \mathbb{E}(B^r) \right] \\
&= \frac{\sigma}{\sigma \left[\frac{\Phi(\mu_1/\sigma)}{\phi(\mu_1/\sigma)} + \frac{\Phi(-\mu_2/\sigma)}{\phi(-\mu_2/\sigma)} \right]} \left[\frac{\Phi(\mu_1/\sigma)}{\phi(\mu_1/\sigma)} \mathbb{E}(A^r) + \frac{\Phi(-\mu_2/\sigma)}{\phi(-\mu_2/\sigma)} \mathbb{E}(B^r) \right] \\
&= \left[\frac{\frac{\Phi(\mu_1/\sigma)}{\phi(\mu_1/\sigma)}}{\left[\frac{\Phi(\mu_1/\sigma)}{\phi(\mu_1/\sigma)} + \frac{\Phi(-\mu_2/\sigma)}{\phi(-\mu_2/\sigma)} \right]} \mathbb{E}(A^r) + \frac{\frac{\Phi(-\mu_2/\sigma)}{\phi(-\mu_2/\sigma)}}{\left[\frac{\Phi(\mu_1/\sigma)}{\phi(\mu_1/\sigma)} + \frac{\Phi(-\mu_2/\sigma)}{\phi(-\mu_2/\sigma)} \right]} \mathbb{E}(B^r) \right] \\
&= (1 - w) \mathbb{E}(A^r) + w \mathbb{E}(B^r)
\end{aligned}$$

where $A \sim TN_+(\mu_1, \sigma^2)$, $B \sim TN_-(\mu_2, \sigma^2)$ and TN_+ and TN_- denote the positively and negatively truncated normal distributions respectively.

If $x \leq 0$ the CDF is given by

$$\begin{aligned}
P(X < x) &= Z^{-1} \int_{-\infty}^x \exp\left[-\frac{1}{2}at^2 + (b + c)t\right] dt \\
&= Z^{-1} \sqrt{2\pi\sigma^2} \exp\left(\frac{\mu_2^2}{2\sigma^2}\right) \int_{-\infty}^x \phi(t; \mu_2, \sigma^2) dt \\
&= Z^{-1} \sqrt{2\pi\sigma^2} \exp\left(\frac{\mu_2^2}{2\sigma^2}\right) \Phi\left(\frac{x - \mu_2}{\sigma}\right) \\
&= \frac{\sigma}{Z} \frac{\Phi\left(\frac{x - \mu_2}{\sigma}\right)}{\phi(-\mu_2/\sigma)} \\
&= w \frac{\Phi\left(\frac{x - \mu_2}{\sigma}\right)}{\Phi(-\mu_2/\sigma)}
\end{aligned}$$

and if $x > 0$ we have

$$\begin{aligned}
P(X < x) &= Z^{-1} \int_{-\infty}^x \exp\left(-\frac{1}{2}at^2 + bt - c|t|\right) dt \\
&= Z^{-1} \sqrt{2\pi\sigma^2} \left[\exp\left(\frac{\mu_1^2}{2\sigma^2}\right) \int_0^x \phi(t; \mu_1, \sigma^2) dt + \exp\left(\frac{\mu_2^2}{2\sigma^2}\right) \Phi\left(\frac{-\mu_2}{\sigma}\right) \right] \\
&= \frac{\sigma}{Z} \left[\frac{\Phi\left(\frac{x-\mu_1}{\sigma}\right) - \Phi\left(\frac{-\mu_1}{\sigma}\right)}{\phi(\mu_1/\sigma)} + \frac{\Phi(-\mu_2/\sigma)}{\phi(-\mu_2/\sigma)} \right] \\
&= 1 - (1-w) \frac{\Phi\left(\frac{\mu_1-x}{\sigma}\right)}{\Phi(\mu_1/\sigma)}.
\end{aligned}$$

For the inverse CDF, we again have two cases. Let $u = P(X < x)$. When

$$u \leq \frac{\sigma}{Z} \frac{\Phi(-\mu_2/\sigma)}{\phi(-\mu_2/\sigma)} = w$$

we solve

$$u = \frac{\sigma \Phi\left(\frac{x-\mu_2}{\sigma}\right)}{Z \phi(-\mu_2/\sigma)}$$

for x to obtain

$$x = \mu_2 + \sigma \Phi^{-1}[(Z/\sigma)\phi(-\mu_2/\sigma)u] = \mu_2 + \sigma \Phi^{-1}[\Phi(-\mu_2/\sigma)u/w].$$

When

$$u > \frac{\sigma}{Z} \frac{\Phi(-\mu_2/\sigma)}{\phi(-\mu_2/\sigma)} = w$$

we need to solve

$$u = 1 - (1-w) \frac{\Phi\left(\frac{\mu_1-x}{\sigma}\right)}{\Phi(\mu_1/\sigma)}$$

for x to obtain

$$x = \mu_1 - \sigma \Phi^{-1}[\Phi(\mu_1/\sigma)(1-u)/(1-w)].$$

The Lasso distribution belongs to the exponential family distributions based on the following in which $\boldsymbol{\theta} = (a, b, c)$ is the set of parameters

$$\begin{aligned}
p(X|\boldsymbol{\theta}) &= h(X) \exp[\boldsymbol{\nu}(\boldsymbol{\theta})\mathbf{T}(X) - \mathbb{A}(\boldsymbol{\theta})] \\
&= \exp\left[-\frac{1}{2}aX^2 + bX - c|X| - \left\{\log\left(\frac{\Phi(\mu_1/\sigma)}{\phi(\mu_1/\sigma)} + \frac{\Phi(-\mu_2/\sigma)}{\phi(-\mu_2/\sigma)}\right) + \log(\sigma)\right\}\right],
\end{aligned}$$

where

$$\begin{aligned}
\boldsymbol{\nu}(\boldsymbol{\theta}) &= (-\frac{1}{2}a, b, -c), \quad h(X) = 1, \quad \mathbf{T}(X) = (X^2, X, |X|), \quad \text{and} \\
\mathbb{A}(\boldsymbol{\theta}) &= \log\left[\frac{\Phi(\mu_1/\sigma)}{\phi(\mu_1/\sigma)} + \frac{\Phi(-\mu_2/\sigma)}{\phi(-\mu_2/\sigma)}\right] + \log(\sigma).
\end{aligned}$$

Mode of the Lasso distribution. Since the normalizing constant does not depend on x , the mode maximizes the log-kernel

$$\ell(x) = -\frac{1}{2}ax^2 + bx - c|x|, \quad a > 0.$$

The function $\ell(x)$ is concave and piecewise quadratic. For $x \geq 0$ we have

$$\ell(x) = -\frac{1}{2}ax^2 + (b-c)x,$$

whose maximizer is $x^* = (b - c)/a$, truncated to the feasible set $x \geq 0$, hence

$$x_+^* = \max\{(b - c)/a, 0\}.$$

For $x \leq 0$ we have

$$\ell(x) = -\frac{1}{2}ax^2 + (b + c)x,$$

whose maximizer is $x^* = (b + c)/a$, truncated to $x \leq 0$, hence

$$x_-^* = \min\{(b + c)/a, 0\}.$$

Comparing the two cases (equivalently, applying the soft-thresholding operator) yields the unique global maximizer

$$\text{mode}(X) = \frac{\text{sign}(b)}{a} \max\{|b| - c, 0\}.$$

In particular, if $|b| \leq c$ then 0 is the mode, while if $|b| > c$ the mode is shifted from b/a toward 0 by c/a .

C Numerical accuracy of Mills ratio approximation

We verified the numerical accuracy of the implemented Mills ratio function by comparing it against a 256-bit arbitrary-precision reference computed using the `Rmpfr` package (?). Over the range $d \in [0, 2000]$, the maximum relative error was 6.31×10^{-14} and the median relative error was 1.25×10^{-14} , corresponding to at least 13 correct significant digits across all tested values (see Table ??).

For comparison, the naive evaluation $\Phi(-d)/\phi(-d)$ using standard double-precision routines (`pnorm/dnorm`) becomes numerically unstable for $d \gtrsim 37.5$, where underflow leads to `NaN` or infinite values. In contrast, the proposed rational approximation remains stable and highly accurate throughout the entire tested range.

Metric	Value
Maximum relative error	6.31×10^{-14}
Median relative error	1.25×10^{-14}
Minimum significant digits	13.20
Median significant digits	13.90
Naive evaluation failure threshold	$d \approx 37.5$

Table 1: Accuracy of the implemented Mills ratio approximation compared to a 256-bit arbitrary-precision reference. Relative errors are computed over selected points in $d \in [0, 2000]$. Significant digits are estimated as $-\log_{10}(\text{relative error})$.